ABDELRAHMAN HASSAN

AI Engineer | LLM, RAG & Agentic Systems | Python Backend Engineer

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PROFESSIONAL SUMMARY

AI Engineer with about 3.5 years of hands-on experience building LLM, RAG, and agentic systems, from LoRA fine-tuning and REST integration to production multi-agent pipelines. Builds reliable Python backends (FastAPI, PostgreSQL, Redis, Celery, Docker) with multi-provider LLM routing and failover (OpenAI, Claude, Gemini), cost control, and automated testing. Delivered freelance GenAI projects for healthcare, travel, education, and SaaS clients, owning work from requirements to release. Studying B.Sc. in Artificial Intelligence at Helwan University.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, C
Agentic AI & LLMs	Multi-Agent Orchestration, LangGraph, LangChain, RAG, Embeddings, Tool/Function Calling, Structured Outputs, Prompt Engineering, LLM Evaluation, Hallucination Auditing, LoRA Fine-Tuning, Local/Quantized LLMs (Ollama, GGUF), LLM Provider Routing and Failover, Cost/Budget Management
Backend & APIs	FastAPI, Flask, Node.js/Express, REST APIs, WebSockets, SSE, Microservices, Celery Background Workers, Token-Bucket Rate Limiting, JWT Authentication
Data & Storage	PostgreSQL, Redis (Lua scripting, caching), SQLAlchemy, Alembic, SQLite, FAISS, ChromaDB, Pinecone, Pandas, NumPy
Frontend	React, Next.js, TypeScript
Testing & DevOps	Playwright (E2E), Bruno (API testing), Docker, Git, Linux, Oracle Cloud, CI/CD Fundamentals, Streamlit
ML Foundations	PyTorch, TensorFlow, scikit-learn, Transformers

PROFESSIONAL EXPERIENCE

AI Engineer (Freelance)

Self-Employed | Cairo, Egypt (Remote) | Feb 2024 - Present

- Led design, development, and deployment of production-grade LLM, RAG, and agentic automation systems for clients in healthcare, travel, education, and SaaS (including Healines, SamaCapeTown, and Erroz Technologies), owning delivery from requirements gathering through release.
- Architected multi-agent workflows and RAG pipelines with LangGraph, LangChain, ChromaDB, FAISS, and Pinecone, using tool calling, structured outputs, contradiction detection, and critic-led hallucination auditing for reliable responses.
- Reduced LLM inference cost and latency by running local 4-bit quantized models (Ollama, GGUF) and implementing cost-aware multi-provider routing with automatic failover across OpenAI, Claude, and Gemini.
- Built containerized backend services with Python, FastAPI, PostgreSQL, and Redis, including atomic Lua-based token-bucket rate limiting, Celery workers, JWT authentication, and Docker; deployed to Oracle Cloud.
- Built full-duplex real-time AI pipelines, including a local speech-to-speech tutoring system (Whisper STT, quantized LLM, RAG, TTS) and a reusable WhatsApp AI Agent SDK with persistent memory and human-in-the-loop controls.
- Applied automated regression testing for AI agents (ReplayLens) and E2E/API testing with Playwright and Bruno.
- Worked directly with founders and non-technical stakeholders to turn business problems into technical specifications and automate manual workflows.

LLM Engineer

Product MasterZ | Egypt (Remote, Contract) | Apr 2023 - Dec 2023

- Worked in a cross-functional engineering team to train and adapt specialized LLMs for internal company domains and business workflows.
- Prepared and curated domain-specific training datasets, ran LoRA fine-tuning and prompt optimization, and evaluated model outputs for accuracy and task alignment.
- Integrated the fine-tuned models into the company backend through REST APIs, enabling inference and automated workflows across internal systems.
- Coordinated asynchronous sprints, dataset reviews, and delivery milestones using Trello, Notion, and Slack.

PROJECTS

Real-Time Arabic AI Voice Tutor | Whisper STT, Quantized LLM, RAG, TTS, React (Freelance, in progress)

- Engineered a full-duplex, local speech-to-speech tutoring pipeline (Whisper STT, quantized 13B LLM, RAG, TTS) with streaming and session memory.
- Built a React frontend with live audio streaming and an AI avatar; decoupled STT, LLM, retrieval, TTS, and UI services for independent scaling.

Multi-Agent Research Automation System | LangGraph, FastAPI, ChromaDB, Ollama (Llama 3.1 8B), Docker
GitHub: github.com/abdelrhmanHasaan/ResearchMultiAgentSystem

- Built a five-agent pipeline (Planner, Scraper, Analyzer, Writer, Critic) that turns a research query into a structured, sourced report (Markdown/PDF).
- Ran fully local with no external API dependency: 4-bit GGUF (Q4_K_M) quantization on about 6 GB VRAM, Nomic embeddings with ChromaDB, served through Dockerized FastAPI microservices.
- Added source-credibility scoring, cross-document contradiction detection, and a critic-led hallucination audit and revision loop.

WhatsApp AI Agent SDK | Python, Node.js, Baileys, WebSockets, Ollama, LLM APIs
GitHub: github.com/abdelrhmanHasaan/wa-agent-sdk

- Built a reusable WhatsApp agent SDK with tool calling, persistent conversation memory, document/image understanding, and multi-agent persona workflows.
- Added cost-aware provider selection, automatic failover, batching/scheduling, human-in-the-loop approval, and safety/rate-limit controls.

SchemaForge - AI Database Architect | React, TypeScript, FastAPI, PostgreSQL, OpenAI, Claude, Gemini
GitHub: github.com/abdelrhmanHasaan/SchemaForge.v2

- Built a natural-language interface that generates PostgreSQL schemas, SQL, relationships, and indexes, with an interactive ER diagram canvas and live schema/SQL synchronization.
- Implemented multi-provider LLM support with streaming chat (SSE), reviewable AI action cards, and one-click transactional deployment to a live database.
- Covered by 53 backend tests and 10 Playwright end-to-end tests.

JobFinder AI Career Assistant | Python, FastAPI, React, TypeScript, LiteLLM, SQLAlchemy, Celery
GitHub: github.com/abdelrhmanHasaan/JobFidner

- Built a career-assistance workflow covering CV parsing, job search and CV-to-job matching, tailored CV generation, and recruiter-email drafting.
- Designed a multi-provider LLM layer with JWT authentication (HttpOnly cookies), role-based admin access, usage metrics, and account data export/deletion.

Devmate | FastAPI, Redis (Lua), PostgreSQL, Alembic, Playwright, Docker
GitHub: github.com/abdelrhmanHasaan/DevMate

- Built a token-bucket rate limiter on Redis using atomic Lua scripts for race-free check-and-decrement under concurrency, and compared it with an application-level implementation for atomicity and performance trade-offs.
- Used Redis for shared state and caching, Alembic for migrations, background workers for async tasks, and Playwright E2E tests to catch UI regressions.

ReplayLens | AI agent testing
GitHub: github.com/abdelrhmanHasaan/ReplayLens

- Record, replay, and regression-test AI agent behavior to catch changes in agent outputs.

Agentic Chatbot Template | FastAPI, PostgreSQL, JWT, SSE

- Reusable chatbot backend with SSE streaming, tool execution, and PDF parsing with structured outputs.

EDUCATION

B.Sc. in Artificial Intelligence - Helwan University, Faculty of Artificial Intelligence | Cairo, Egypt | In progress

Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures and Algorithms, Databases, Cloud Computing, LLMs

CERTIFICATIONS & ACTIVITIES

- IBM AI/ML Professional Certificates (Coursera): Machine Learning, Deep Learning, Generative AI, NLP
- NVIDIA: AI Infrastructure and Operations Fundamentals (Coursera)
- NCPC: Nile University National Problem-Solving Competition, Participant (2024)

LANGUAGES

Arabic (Native) | English (Written: professional; Spoken: developing)